

Web Technology



React Bootstrap

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Introduction - Bootstrap

1. What is Bootstrap?

Bootstrap is an open-source front-end framework containing ready-made CSS classes and JavaScript components for building responsive websites and web applications.

Introduction - Bootstrap

1. What is Bootstrap?

- **Framework CSS**
- **By Twitter – Opensource**
- **Built-in Themes**
- **Custom Themes**
- **Aimed – SPAs and Mobile Apps**
- **Depends on JQuery** (older Bootstrap versions were, but modern Bootstrap is not.)
- **Responsive**
- https://www.w3schools.com/whatis/tryit.asp?filename=trywhatis_bootstrap&stacked=h

Introduction - React Bootstrap

1. What is React Bootstrap?

- **Reimplementation of Bootstrap – React Component**
- **No JQuery or Javascript Dependency**
- **Renders React Components**

React Bootstrap is a UI framework that provides **Bootstrap components as React components**. It allows developers to build responsive and attractive web interfaces using Bootstrap's styling while following React's component-based approach.

React Bootstrap - Installation

1. Three ways

- **Bootstrap Content Delivery Network (CDN)**
- **Installing and Importing Bootstrap in React**
- **Installing react-bootstrap**

React Bootstrap - Installation

1. **Three ways**
 - **Bootstrap Content Delivery Network (CDN)**
 - **Installing and Importing Bootstrap in React**
 - **Installing react-bootstrap**
2. **Use react-bootstrap**

React Bootstrap - Installation

1. Three ways

- **Bootstrap Content Delivery Network (CDN)**
- **Installing and Importing Bootstrap in React**
- **Installing react-bootstrap or reactstrap**

2. Use react-bootstrap

```
npm install react-bootstrap bootstrap
```

React Bootstrap - Importing

1. From folder
2. Using {}
3. Default vs Multiple Components

App.jsx

```
import Button from 'react-bootstrap/Button';  
import Alert from 'react-bootstrap/Alert';  
import Card from 'react-bootstrap/Card';  
import Form from 'react-bootstrap/Form';  
import Navbar from 'react-bootstrap/Navbar';  
import Nav from 'react-bootstrap/Nav';  
import Container from 'react-bootstrap/Container';  
import Modal from 'react-bootstrap/Modal';  
import Pagination from 'react-bootstrap/Pagination';  
import Offcanvas from 'react-bootstrap/Offcanvas';
```


React Bootstrap - Example

```
<Stack direction="horizontal" gap={2}>  
  <Button as="a" variant="primary">  
    | Button as link  
  </Button>  
  <Button as="a" variant="success">  
    Button as link  
  </Button>  
</Stack>;
```

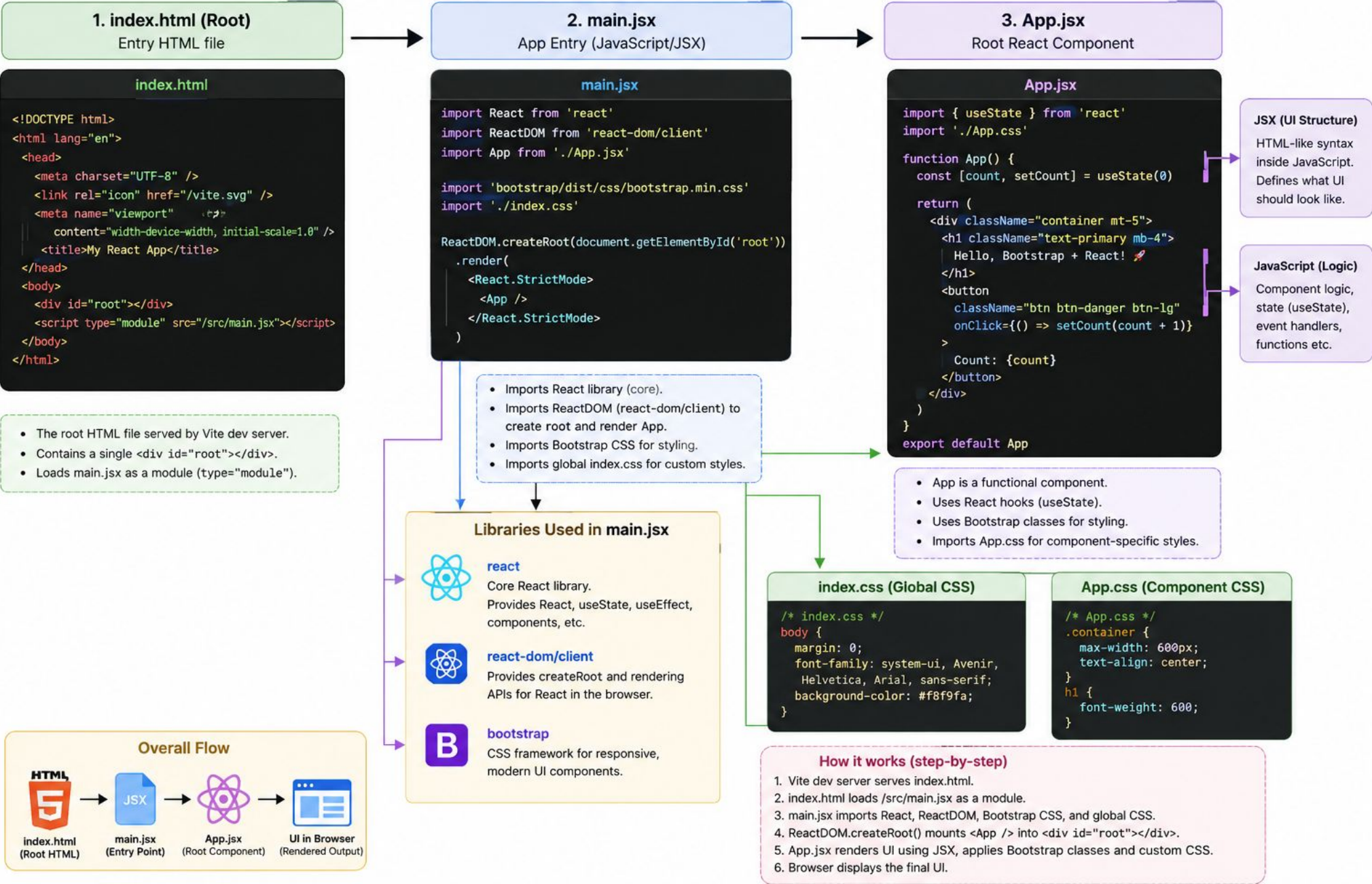
React Bootstrap - Example

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```

Button as link

Button as link

React (Vite) Project Structure & Flow



React Bootstrap - Example

```
npm create vite@latest myapp -- --template react
cd myapp
npm install
npm install react-bootstrap bootstrap
npm run dev
```

See the default vite page

replace App.jsx and main.jsx in ./src folder

Syllabus

UNIT 1 - WEB SITE BASICS

9

Web Essentials: Introduction to HTML, CSS, JavaScript - Data types, Arrays, Functions, Other built-in objects. DOM - Document tree traversal and manipulations, Event handling. Introduction to AJAX - Request, Response.

UNIT 2 - CLIENT-SIDE PROGRAMMING

9

Frontend Development: React JS/ React Overview, Model-View-Controller (MVC) Architecture, Introduction to ECMAScript6 (ES6), Virtual DOM, Javascript with XML (JSX), Components - Functional Component, Exporting and Importing Components, States, Props, Hooks – useState, useEffect, React Routing, Bootstrap.

Summary

- 1. Introduction – Bootstrap**
- 2. Introduction – React Bootstrap**
- 3. React Bootstrap – Installation**
- 4. React Bootstrap – Example**
- 5. Syllabus - Chapter - 1 and 2**

References

1. Robin Wieruch, “The Road to React: The ReactJS with Hooks”, Lean Publishing, 2023 Edition.

2. Online Sources

- [Introduction | React Bootstrap \(react-bootstrap.netlify.app\)](https://react-bootstrap.netlify.app)
- [Using Bootstrap with React: Tutorial with examples - LogRocket Blog](#)
- [Understanding React-Bootstrap and Making It Work for You | Pluralsight](#)

THANK YOU

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